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Literature Review Instructions

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1 Abstract

Concise summary of purpose, methods, key results, and contribution.

Highlight uniqueness: “linking TESM with real-estate listings.”

2 Introduction & Motivation

Context: TESM, crop suitability, and property market dynamics.

The Tasmanian Enterprise Suitability Map (TESM) is a geospatial decision-support tool that integrates soil, climate, and landscape data to classify land as well suited, suitable, marginally suitable, or unsuitable for a range of agricultural enterprises. Developed by the Tasmanian Institute of Agriculture (TIA) in collaboration with industry, TESM supports planning and investment by providing science-based assessments of crop potential across Tasmania. By making suitability information publicly accessible through LISTmap, it has become a cornerstone of agricultural planning and resource allocation in the state.

Motivation: current LISTMap/real-estate platforms are passive & fragmented.

Despite its strengths, existing platforms present two key limitations. First, they respond poorly to changes in agribusiness trends, including the emergence of new crops that may become economically viable or strategically important for food security in the near future. Second, they are not directly linked with the property market, leaving a critical gap between geospatial crop suitability analysis and the practical realities of land acquisition. Prospective buyers or agribusiness investors seeking to identify land parcels that align with their crop preferences must manually consult multiple sources, including property listing sites, agronomy reports, and local council data. This process is time-consuming, fragmented, and often discourages comprehensive due diligence.

Prospective buyers and landholders in Tasmania therefore lack an efficient mechanism to assess which crops are suitable for newly listed properties at the time they appear on the market. Current real-estate platforms do not integrate geospatial factors such as soils, climate, topography, or water access into actionable summaries, meaning that opportunities are frequently missed. This project addresses that gap by automatically assessing crop suitability for new property sales listings and directly linking users with properties that match their preferred crops. To achieve this, the project develops an integrated system that combines property listing information, cadastral parcel boundaries, and statewide crop suitability layers. This system computes parcel-level suitability and presents the results through an interactive Shiny dashboard, providing users with a more personalised and efficient way to evaluate investment opportunities.

Building on this problem statement, the present study redefines the role of TESM by moving beyond the static functions of LISTmap. The project develops an integrated system that links live property sales data with crop suitability assessments derived from TESM, providing users with a more personalised and efficient way to identify land that matches their preferred crops. By combining property listing information, cadastral parcel boundaries, and statewide crop suitability layers, the system computes parcel-level suitability and delivers a more dynamic and actionable decision-support tool.

The significance of this research extends across several domains. For investors, the system enables better decision-making by reducing business risk and enhancing opportunity discovery. It assists those who already have specific crop strategies by identifying parcels that match their requirements as soon as they are listed, while also supporting landholders or capital investors who wish to explore which crops are most suitable for a given parcel. For the **agribusiness sector**, it provides a scalable framework for integrating suitability analysis into real-time land acquisition decisions, enabling faster and more confident responses in an environment of rising land values, shifting crop demands, and climate uncertainty. By reducing the risk of costly misallocation, the system ensures that capital and resources are directed toward opportunities with the greatest long-term viability. For **policy and planning bodies**, the system demonstrates how public geospatial resources such as TESM can be repurposed into practical, user-focused applications that bridge information gaps between government datasets and private-sector decision-making. It also highlights that these resources are not confined to a small niche of agricultural specialists, but can deliver value to mainstream business communities, including property-based investors and developers. This broader relevance strengthens the case for continued improvement of TESM and LISTmap through the inclusion of additional variables and more sophisticated modelling approaches. By connecting suitability assessment with property market dynamics, this project contributes to the creation of a more responsive, transparent, and data-driven agricultural investment environment in Tasmania.

3 Objective

This project aims to integrate TESM crop suitability layers with live property listing data in order to generate parcel-level suitability assessments. Building on this foundation, it develops an interactive Shiny dashboard that allows users to view ranked property recommendations, examine limiting factors, and directly access the corresponding property listings. The project also evaluates the tool's functionality through case studies in Tasmania and explores its potential for broader application, extending to regions with comparable suitability datasets and to future industry-facing implementations.

4 Literature Review

4.1 The Tasmanian Enterprise Suitability Map (TESM)

TESM integrates high-resolution soil, climate, and topographic datasets using deterministic, rule-based thresholds to classify land across Tasmania into categories ranging from well suited to unsuitable. Its modelling combines digital soil mapping (DSM), regional climate models, and terrain attributes, resampled to raster layers at 30–80 m resolution. Each enterprise is defined by a crop-specific rule set that specifies acceptable ranges for soil chemistry, drainage, slope, and climatic parameters such as frost risk, chill hours, and rainfall. The outcome is a categorical map where limiting factors are highlighted, allowing users to identify both opportunities and constraints (Kidd et al., 2012; Kidd et al., 2014; Webb et al., 2014).

One of TESM's main strengths lies in its transparency. Because thresholds are rule-based and expert-defined, stakeholders can trace why a parcel is classified as suitable or unsuitable, reinforcing interpretability and trust. Its incorporation of climate change projections for 2030 and 2050 (RCP 8.5) further extends its value by enabling forward-looking planning. In addition, the Enterprise Versatility Index (EVI) aggregates suitability scores across multiple crops, helping identify parcels capable of supporting diverse enterprises. These features make TESM a cornerstone of agricultural planning in Tasmania.

Despite these advantages, the literature highlights several persistent limitations. A central issue is the assumption that irrigation is universally available. While designed to reflect Tasmania's expanding irrigation infrastructure, this assumption obscures the true costs of water access and masks risks for investors. Equally important is TESM's reliance on a deterministic framework: small deviations beyond a threshold can disproportionately alter suitability classifications, even when agronomic implications may be minor. This rigid structure contrasts with probabilistic approaches that incorporate uncertainty estimates and confidence levels, as used in tools such as New Zealand's S-Map or FAO's GAEZ framework.

Table 1: *List of All Current Crops in TESM*

Type	Crops
cereals	barley, linseed, wheat
perennial horticulture	blueberriesNHB, blueberriesSHB, cherries, hazelnuts, olives, raspberries, sparklingwg, strawberries, tablewg
vegetables	carrots, carrotseed, onions, potatoes
pharmaceuticals	hemp, poppies, pyrethrum

pastures	cocksfootcontinental, cocksfootmediterranean, lucerne, phalaris, redclover, ryegrass, strawberryclover, tallfescuecontinental, tallfescuemediterranean, whiteclover
forestry	E_globulus_tree, E_nitens_tree, P_radiata_tree

Another limitation is the restricted flexibility for new or specialty crops. TESM currently covers around 33 enterprises, but adding further crops requires formal updates by custodians at the Department of Natural Resources and Environment Tasmania. This process slows responsiveness to emerging commodities or food-security priorities. Crop scoring rules also lack standardisation: some enterprises use broad integer scales, while others apply finer increments. This inconsistency complicates cross-crop comparisons and undermines interpretability. Moreover, the depth of crop-specific modelling varies; while some crops emphasise a single dominant climatic factor, others require multiple interacting variables that are not adequately captured in the current weighting scheme.

Limitations also extend to the delivery platform. TESM is primarily accessed through LISTmap, an Esri-based GIS interface. While users can query, zoom, and overlay additional layers, integration remains largely static and technical. Adding new crop layers requires manual preparation in GIS software and approval by custodians, creating barriers for non-specialist users. LISTmap also lacks collaborative features, preventing users from contributing or sharing crop-specific data. This makes the system slow to adapt and discourages widespread engagement.

The absence of uncertainty estimates is another recurring critique. By treating thresholds as fixed, TESM obscures the variability inherent in soil and climate predictions and overlooks differences in management practices such as irrigation scheduling, pest control, or crop rotations. These management factors can significantly affect yield outcomes but are not reflected in the current suitability framework. Scholars have also emphasised the challenges of mapping niche crops in Tasmania. Specialty commodities often lack locally validated thresholds or long-term datasets, meaning suitability rules must be extrapolated from limited evidence. Without robust ground-truthing and monitoring, these layers are less reliable than those for mainstream crops (Magliocca et al., 2020; Pramanik et al., 2023).

Together, these limitations indicate that while TESM is a robust first-pass planning tool, it underrepresents both uncertainty and market adaptability. This has important consequences for its use in dynamic contexts such as real-estate transactions and agribusiness investment, where responsiveness and reliability are crucial.

4.2 Crop Suitability Mapping Approaches in Broader Context

The broader literature on crop suitability mapping highlights a variety of approaches that go beyond deterministic rule sets. The FAO framework, for example, conceptualises land evaluation as a match between land characteristics and crop requirements, but more recent implementations incorporate probabilistic modelling and scenario analysis. The Global Agro-Ecological Zones (GAEZ) project uses both agro-climatic indices and crop simulation models to provide suitability estimates across scales. Similarly, New Zealand's S-Map integrates uncertainty estimates into soil and suitability products, allowing users to consider confidence levels alongside categorical ratings.

Table 2: *List of All Variables in TESM*

Class	Variables
climate	Air temperature, Chill hours, Daily maximum temperature, Extreme heat risk, Frost risk, Growing Degree Days, Growing Season Temperature, Heat stress, Hot weather during summer, Rainfall
soil attribute	Depth to sodic layer, Duplex soil, Electrical conductivity, Exchangeable calcium, Exchangeable magnesium, Soil depth, Soil Depth, Soil drainage, Soil texture, Stone abundance
topography	Elevation, Slope

Compared to these approaches, TESM's deterministic framework has advantages in transparency and simplicity, but disadvantages in its inability to propagate uncertainty or reflect management practices. Probabilistic approaches, as argued by Magliocca et al. (2020) and Pramanik et al. (2023), better represent the variability of real-world farming conditions, such as microclimate effects, adaptive irrigation, or pest management. Furthermore, models that integrate socio-economic variables — such as proximity to infrastructure, cost of water access, or land tenure — provide a more holistic picture of suitability. These elements are largely absent in TESM.

Research also emphasises the challenge of mapping niche or specialty crops. Unlike cereals or horticultural staples, specialty crops often lack long-term, high-resolution environmental datasets. Locally validated crop-specific thresholds are rare, making rule-based suitability assessment difficult. This is a key limitation for Tasmania, where diversification into specialty commodities (e.g., pharmaceutical crops, novel horticulture) is increasingly important.

Thus, while TESM remains state-of-the-art in terms of regional agricultural planning, it is not as sophisticated as some global frameworks. The literature underscores the need for integrating un-

certainty, management practices, and niche crop datasets into suitability mapping to ensure greater robustness and flexibility.

4.3 Real-Estate Listing Platforms and Agricultural Land Information

Parallel to crop suitability mapping, real-estate platforms provide a different but equally crucial dimension of land information. Mainstream platforms such as Domain.com.au and RealEstate.com.au emphasise standard property attributes: area, boundaries, tenure, buildings, proximity to schools and transport, and neighbourhood features. While these characteristics are useful for urban or residential buyers, they are insufficient for agribusiness investors, as they provide no objective information about soil, climate, or agricultural suitability.

Niche platforms such as Farmbuy.com and Farmproperty.com.au cater to rural buyers but still fall short of offering integrated agronomic assessments. At best, they allow sellers to manually list “suitable crops” in descriptions, but these claims are unverified and subjective. There is no systematic use of climate or soil datasets. This reliance on user-provided detail introduces bias and raises reliability concerns.

More broadly, the property technology or *proptech* sector has advanced in areas such as automated valuation models, neighbourhood analytics, and energy efficiency scoring. However, integration with agricultural geospatial data remains absent. As a result, prospective agribusiness investors face a fragmented process: manually cross-checking listings against external maps, reports, and local council data. This process is inefficient, error-prone, and inaccessible to those without technical GIS expertise.

The literature highlights similar gaps internationally. Agricultural investors frequently lack tools that integrate biophysical data with market availability, creating barriers to efficient and sustainable land-use decisions. This has motivated calls for more integrated land information systems that bridge the property market and agricultural planning domains.

4.4 The Integration Gap

The synthesis of the above strands reveals a clear gap. On one side, TESM and similar frameworks provide scientifically robust crop suitability assessments, but are delivered through GIS-centric platforms (e.g., LISTmap) that are passive, inflexible, and poorly connected to real-time markets. On the other side, property listing platforms are dynamic, widely used, and commercially oriented, but ignore biophysical suitability altogether.

No existing system links these two domains into an actionable decision-support interface. Even niche agricultural listing sites rely on subjective seller descriptions, while suitability maps remain siloed

in specialist GIS platforms. This disconnect prevents timely, evidence-based decisions by investors, farmers, and agribusinesses.

Addressing this integration gap is the unique contribution of the present project. By combining TESM crop suitability layers with live property sales data, and presenting the results through a Shiny dashboard, the project creates the first prototype of a tool that makes suitability analysis actionable at the point of market transaction. This directly responds to the limitations identified in both TESM and real-estate platforms, advancing the literature by demonstrating how geospatial agronomic data can be embedded in everyday property decision-making.

TESM data can be accessed and interacted with via [LISTmap](#), where users can explore suitability ratings for different crops in specific locations and understand the limitations or manageable constraints of the land. Users can view and query the maps, zoom into specific locations, and overlay other relevant environmental or administrative spatial layers. Access to the suitability maps, crop rules and Enterprise fact sheets and user guides are provided on the [Enterprise Suitability Maps](#) page. ‘Overall, TESM provides a science-based, spatially detailed guide to support sustainable agricultural development across Tasmania.

4.5 Critically assess limitations

According to the each crop rules dataset, extracted from the [website](#), each crops has different combination of crucial climate variables and topography to consider. This claim supported by [this data](#) from Tasmanian Government. This makes modelling an accurate and reliable model for all crops challenging. Missing data happen in all variables.

Adding on, by investigating the crops rules dataset (*see Appendix Table A1*), there are limitations of the lack of common/standardize variables. This makes the mapping harder, since the lack of common variables means the map is less reliable.

4.6 Suggested Comparator Models

New Zealand S-map & Crop Suitability Layers

New Zealand’s national S-map integrates soil, climate, and terrain data at a 500 m resolution to assess crop suitability, including specialty crops such as truffles, hops, and few specialty wine grapes. It is jointly maintained by Plant & Food Research, [Manaaki Whenua Landcare Research](#), [NIWA](#), and [MPI](#). Compared to Tasmania’s TESM, the framework includes more explicit representation of biophysical constraints and yields expressed as a range, offering users insight into uncertainty.

- Inputs: Soil (depth, pH, drainage from S-map), climate (temperature, rainfall from NIWA Virtual Climate Station), and terrain (slope, aspect).

- **Methods:** Rule-based GIS modelling adapted from Kidd et al. (2015), similar to modelling reference of TESM, with suitability defined in four categories (well-suited, suitable, marginal, unsuitable) based on most-limiting factor logic. Rules are continuously refined with expert feedback.
- **Uncertainty:** Yield estimates are presented as lower and upper bounds rather than single values, acknowledging inherent variability. Suitability classes carry a “moderate reliability” disclaimer at the national scale, since estimates depend on expert judgment, national yield statistics, and generalised soil–climate interactions.
- **Extensibility:** Modular design allows addition of new crops and explicit management scenarios (e.g. irrigation, liming, drainage) demonstrate adaptability to different farm practices.

Compared to TESM, the New Zealand’s S-map framework provides national-scale consistency and explicit yield ranges, though with less resolution for farm-level decisions.

FAO GAEZ: Agro-MAPS

The FAO Global Agro-Ecological Zones (GAEZ) framework, with its [Agro-MAPS](#) database, is a global land evaluation system designed to assess agricultural potential under current and future climate conditions. Unlike Tasmania’s TESM, which are state-level, GAEZ integrates global datasets on soils, climate, terrain, and agricultural statistics to evaluate suitability, potential yields, and yield gaps across multiple scales.

- **Inputs:** Climate (historical and scenario-based, RCP futures), soils and terrain (from the Harmonized World Soil Database and SRTM slope data), land cover/use, phenology and crop calendars, population and livestock distributions, and national/subnational agricultural statistics.
- **Methods:** Multi-module modelling system. Module I analyses agro-climatic variables (length of growing period, water balance, frost days); Module II calculates biomass and yields; Modules III–IV assess agro-climatic and edaphic (soil/terrain) constraints; Modules V–VII integrate evaluations, downscale production statistics, and compute yield gaps. Suitability is classed by limiting factors across irrigated conditions, with input/management levels considered.
- **Uncertainty:** Addressed by scenario modelling (RCP climate pathways, CO₂ fertilisation effects) and by providing ranges for yields and constraints. Downscaling of agricultural statistics incorporates calibration with national data.
- **Extensibility:** Highly flexible. GAEZ accommodates multiple management levels, irrigation systems, and climate scenarios, and is updated iteratively (GAEZ v5 is underway). Crop sets can

expand, and the framework links to biodiversity, protected areas, and socio-economic overlays for policy support.

Compared to TЕСM, GAEZ provides much broader geographic scope and scenario-based foresight, but at coarser resolution and less direct farm-scale applicability.

Comparator Summary

	Tasmania	NZ S-map & Crop		
Feature	TESM	Layers	FAO Agro-MAPS	QLD LSG
Inputs	Soil, climate, topography from state datasets; assumes irrigation is available.	Soil attributes (S-map), climate (NIWA Virtual Climate Station), terrain; crop-specific rules.	Global climate (historical + scenarios), soils (HWSD), terrain (SRTM), land cover/use, crop calendars, ag. statistics (Agro-MAPS).	Soil, climate, topography; broader hazards including irrigation performance, acid sulfate soils, pests, salinity.
Methods	Rule-based, deterministic thresholds; “most limiting factor” approach; outputs 9 suitability classes (with management options).	Rule-based GIS modelling, 4 suitability classes; expert-reviewed iterations; yields expressed as ranges.	Multi-module system (Modules I–VII): agro-climatic analysis, yield modelling, constraints, soil/terrain suitability, integration with national stats, yield gap analysis.	FAO land evaluation framework adapted for Queensland; practitioners build tailored rule-sets; applies most limiting factor principle.
Uncertainty	Not explicitly quantified; only “manageable constraints” noted.	Yields given as lower/upper ranges; categorical maps rated “moderate reliability” at national scale.	Explicitly incorporates climate scenarios, CO2 effects, yield ranges, and calibration with national statistics.	Not directly quantified; comprehensiveness comes from inclusion of wider hazard classes.

	Tasmania	NZ S-map & Crop		
Feature	TESM	Layers	FAO Agro-MAPS	QLD LSG
Extensibility	Crop list fixed but new crops added via rule-sets and consultation; designed for state planning.	Modular; new crops/scenarios can be added; includes specialty crops (e.g. truffles) and management scenarios (irrigation, drainage, liming).	Highly extensible: multiple management levels, irrigated vs rain-fed, climate change scenarios; regularly updated (v5 in development).	Flexible framework; rule-sets built per crop/region by practitioners; applied across regional planning and impact assessment.
Scale/Resolution	State of Tasmania; fine resolution (30–80 m); farm to regional planning.	National (NZ); 500 m resolution; suitable for regional benchmarking, not block-level.	Global; coarse (~5 arc-min, ~10 km); intended for national, regional, and global planning, not local farm scale.	Queensland-wide; applied for regional planning, EIA, irrigation development, and farmland protection.

5 Data Sources

Domain.com.au for property listings (criteria: TAS, $\geq 10\text{Ha}$).

Cadaster (parcel boundaries).

TESM crop suitability rasters (local vs WMS).

Address datasets (for matching).

Optional: subscriber/user preference logic.

Include a table summarising sources (format, size, relevance).

6 Methodology

6.1 Data Pipeline

Step-by-step: scraping $\rightarrow$ address matching $\rightarrow$ parcel overlay $\rightarrow$ suitability calculation $\rightarrow$ dashboard.

Insert diagram (supervisor specifically asked).

Mention database structure (DuckDB or equivalent).

6.2 Suitability Computation

How these crops are determined suitable

Tasmania's Enterprise Suitability Maps (TESM) are modelled using a deterministic, rule-based framework grounded in digital soil assessment (DSA). The soil component relies on machine learning (interpolation) and geostatistical methods to predict soil attributes from field observations and environmental covariates while quantifying uncertainty. Climate and terrain factors are integrated through threshold rules derived from agronomic literature, expert workshops, and industry consultation.

The process uses a "most-limiting-factor" approach. This means that, even if most factors are excellent, just one poor factor can determine the overall suitability. In other words, the weakest link sets the limit:

Suitability levels are usually scored or classified from:

- **1.0 Well suited:** All individual criteria are "Well suited."
- **2.0 Suitable:** "The lowest category among the criteria is Suitable, with all others rated higher."
- **3.0 Moderately suitable:** One or more "Moderately suitable" factors.
- **4.0 Unsuitable:** Any factor is "Unsuitable" (or the land is protected, urban, or open water).

If a location scores “Well suited” for everything except drainage (which is “Unsuitable”), the whole site is deemed “Unsuitable” for that crop, because the poor drainage is expected to severely limit success.

Improving Suitability - If a limiting factor is considered manageable (a “manageable constraint”), landowners can take steps to address it. For example:

- **Low soil pH** : Applying lime can raise pH, making the soil suitable for crops like barley.
- **Poor drainage** : Installing drainage systems can help.
- **Excess stones**: Stones can be mechanically or manually removed.

If such mitigations are feasible, the suitability rating may improve from “Moderately suitable” to “Suitable (with management).”

Suitability is determined by rating all important growing conditions, then choosing the lowest rating among these (the most limiting factor) as the overall score. If that limiting factor can be addressed by practical management techniques, the land’s suitability rating for a crop might be increased to reflect that potential improvement.

In summary, TESM combine expert-defined deterministic rules with advanced digital soil mapping to provide a practical, interpretable, and scientifically robust tool for land-use planning, acknowledging inherent limitations in data and modelling assumptions (Kidd et al., 2012; Kidd et al., 2014; Webb et al., 2014).

Majority pixel rule (explain rationale of the $\geq 50\%$).

Alternative approaches (weighted average, thresholds).

6.3 Dashboard Development

Framework: R Shiny.

Inputs (crop preference, filters).

Outputs (listing table, map overlay, “go to website” button).

7 Results

7.1 Functionality Showcase

Show examples of scraped listings.

Overlay results (parcel + TESM suitability).

Dashboard screenshots.

7.2 Case Example(s)

One or two properties as illustrations: “This property in NW Tasmania is 80% suitable for barley, 60% for carrots.”

8 Discussion

Strengths: first tool combining TESM with live property data.

Challenges:

Address matching issues.

Local rasters vs WMS (computational trade-off).

Crop coverage (cherries/hazelnuts missing).

Opportunities for improvement:

Add more crops.

Automate subscriber notifications.

Link climate projections.

9 Conclusion

Restate contribution: bridging TESM with real-estate.

Highlight impact: empowers faster, data-driven agribusiness investment.

Point to future expansion: broader crops, live updates, policy use.

10 Resources

- TESM: <https://nre.tas.gov.au/agriculture/investing-in-irrigation/enterprise-suitability-toolkit/enterprise-suitability-mapsy>
- TESM Modelling References:
- NZ S-map: <https://smap.landcareresearch.co.nz/>
- FAO EcoCrop: <https://ecocrop.fao.org/>